M&AE 4070/5070 - Dynamics of Flight Vehicles

Case Study 3 Solutions

Trimming a stable aircraft - Horizontal lifting surfaces

The objective of this case study is is to understand how the horizontal lifting surfaces of a classical aircraft (main wings + empennage) work. In particular, we want to figure out how the local front and rear C_z are distributed in order to satisfy $\boxed{\mathbf{M}}$ at steady state. The results would then typically be used in the aircraft design process, to evaluate the size of the empennage. They also provide a picture of what happens from the aerodynamic point of view when nearing stall conditions, which is important for safety reasons. All relevant formulas can be found in Chapter 3 lecture notes.

Pick your favorite long-haul conventional aircraft and look up approximate geometric information for it (aspect ratios, surface areas of wind and empennage, distance between wing and empennage, mean aerodynamic chord etc.) Assume that a high C_z (e.g. when getting close to landing) is around 1, while $C_{m_0} \approx -0.1$.

1. Determine an appropriate cruise C_z for your aircraft.

   Solutions. Commercial aircrafts typically aim to maximize their range. The expression to calculate range involve the ratio $\frac{\sqrt{C_z}}{C_x}$, and range will be maximum if that ratio is maximal. Maximizing $\frac{\sqrt{C_z}}{C_x}$ is equivalent to minimize the inverse ratio: $\frac{C_x}{\sqrt{C_z}}$. Let's call A this ratio:

   $$A = \frac{C_x}{\sqrt{C_z}} = \frac{Cx_0}{\sqrt{C_z}} + \frac{C_z^{3/2}}{\pi \mathbf{AR}} \tag{1}$$

   A is a function of C_z. We want to find the C_z^{opt} that minimizes A, which is found by differentiating A with respect to C_z and setting the result to 0. Strictly speaking, that would only indicate a local extremum. But a quick look at the function itself shows that this extremum is indeed a global minimum.

   $$\frac{dA}{dC_z} = -\frac{1}{2} C_{x_0} C_z^{-3/2} + \frac{3}{2} \frac{C_z^{1/2}}{\pi \mathbf{AR}} = 0 \tag{2}$$

   $$C_z^{\text{opt}} = \sqrt{\frac{1}{3} \pi \mathbf{AR} C_{x_0}} \tag{3}$$

   Taking a typical aspect ratio of $\mathbf{AR} = 8$ and a typical drag coefficient at zero lift of 0.02, we obtain:

   $$C_z^{\text{opt}} = \sqrt{\frac{8\pi 0.02}{3}} = 0.41 \approx 0.4 \tag{4}$$

2. Estimate C_{z_W} and C_{z_E} for a static margin of 10% and a static margin of 30%, and for cruise and high C_z values. Comment on your results. In particular, do you think the empennage could stall before the main wing when the pilot slowly increase the aircraft angle of attack?

   Solutions. For this question, we use the following geometric parameters, which are representative of a medium-sized aircraft:

   - Ratio of surface areas: $\frac{S_E}{S_W} = 0.255$.
   - Mean aerodynamic chord: $\ell = 6.6$ m.
   - Ratio between reference length ℓ and distance between wing and empennage d: $\frac{\ell}{d} = 0.33$.

- $C_{m_0} = -0.1$.
- Aspect ratios: $\mathbf{AR}_W = 8$ and $\mathbf{AR}_E = 5$.

To evaluate how C_{z_E} varies as a function of C_{z_W}, we use the following two results:

$$S_W C_{z_W} + S_E C_{z_E} = S C_z \tag{5}$$

with $S = S_W$, and

$$\frac{C_{z_W}}{C_{z_E}} = \frac{S_E}{S_W} \frac{1 + (R+1)B}{R - (R+1)B} \tag{6}$$

with

$$R = -\frac{\ell_{WF}}{\ell_{EF}} = \frac{S_E}{S_W} \frac{1 - 2/\mathbf{AR}_W}{1 + 2/\mathbf{AR}_E} = 0.255 \times 0.54 = 0.137 \tag{7}$$

and

$$B = -\frac{\ell}{d}\left(m_s + \frac{C_{m_0}}{C_z}\right) \tag{8}$$

For a static margin of 10% (*ie* $m_s = -0.1$):

C_z	B	C_{z_W}	C_{z_E}
0.4	0.116	0.4	0.007
1	0.066	0.95	0.21

and for a more negative static margin, say 30% (*ie* $m_s = -0.3$):

C_z	B	C_{z_W}	C_{z_E}
0.4	0.18	0.43	-0.096
1	0.132	1.01	-0.046

For a value of the static margin within the specified range, C_{z_E} increases when C_{z_W} increases, that is:

$$\frac{\partial C_{z_E}}{\partial C_z} > 0 \tag{9}$$

But when looking at actual values, because of the deflection created by the main wing which reduces the angle of attack felt by the empennage, C_{z_E} is much smaller than C_{z_W}. if we assume the same $C_{z_{\max}}$ for both wing and empennage, it is therefore not possible for the empennage to stall before the main wing.

3. Find where the aerodynamic center of your aircraft is at cruise conditions. Where is it located relative to the aerodynamic center of the main wing?

Solutions. For a typical transport aircraft, the horizontal rear empennage results in an aircraft aerodynamic center located behind that of the wing. We can evaluate the distance between the wing aerodynamic center and the aircraft aerodynamic center, ℓ_{WF}, that is, we can find where the aerodynamic center is on that aircraft using the following 2 equations:

$$\frac{\ell_{WA}}{\ell_{EA}} = -\frac{S_E}{S_W} \frac{1 - 2/\mathbf{AR}_W}{1 + 2/\mathbf{AR}_E} = -R \tag{10}$$

and

$$\ell_{WE} = d = \ell_{WA} - \ell_{EA} \tag{11}$$

Introducing one into the other, we obtain:

$$\frac{\ell_{WA}}{\ell} = \frac{x_W - x_A}{\ell} = \frac{d}{\ell}\frac{R}{1+R} = 3.03\frac{0.14}{1.14} = 37\% \tag{12}$$

The empennage on a typical transport aircraft pushes back the aerodynamic center of the aircraft behind the wing aerodynamic center by a distance equal to 37% the mean aero chord.